Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

```
R1#show ip route ospf
O     192.168.2.0 [110/65] via 192.168.12.2, 00:10:55, Serial0/0/0
O     192.168.3.0 [110/65] via 192.168.13.2, 00:08:05, Serial0/0/1
      192.168.23.0/30 is subnetted, 1 subnets
O        192.168.23.0 [110/128] via 192.168.12.2, 00:07:22, Serial0/0/0
                      [110/128] via 192.168.13.2, 00:07:22, Serial0/0/1
```

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     1.0.0.0/32 is subnetted, 1 subnets
C       1.1.1.1/32 is directly connected, Loopback0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
O    192.168.2.0/24 [110/65] via 192.168.12.2, 00:00:35, Serial0/0/0
O    192.168.3.0/24 [110/65] via 192.168.13.2, 00:19:18, Serial0/0/1
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/30 is directly connected, Serial0/0/0
L       192.168.12.1/32 is directly connected, Serial0/0/0
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/30 is directly connected, Serial0/0/1
L       192.168.13.1/32 is directly connected, Serial0/0/1
     192.168.23.0/30 is subnetted, 1 subnets
O       192.168.23.0/30 [110/128] via 192.168.13.2, 00:00:35, Serial0/0/1
                        [110/128] via 192.168.12.2, 00:00:35, Serial0/0/0

R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
22.22.22.22       0   FULL/  -        00:00:32    192.168.12.2    Serial0/0/0
33.33.33.33       0   FULL/  -        00:00:32    192.168.13.2    Serial0/0/1
```

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     3.0.0.0/32 is subnetted, 1 subnets
C       3.3.3.3/32 is directly connected, Loopback0
O    192.168.1.0/24 [110/65] via 192.168.13.1, 00:19:54, Serial0/0/0
O    192.168.2.0/24 [110/129] via 192.168.13.1, 00:01:16, Serial0/0/0
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0
     192.168.12.0/30 is subnetted, 1 subnets
O       192.168.12.0/30 [110/128] via 192.168.13.1, 00:04:26, Serial0/0/0
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/30 is directly connected, Serial0/0/0
L       192.168.13.2/32 is directly connected, Serial0/0/0
     192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.23.0/30 is directly connected, Serial0/0/1
L       192.168.23.2/32 is directly connected, Serial0/0/1

R3#
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
11.11.11.11       0   FULL/  -        00:00:34    192.168.13.1    Serial0/0/0
```

What interface is R3 using to route to the 192.168.2.0/24 network? **s0/0/0**

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? **129**

Does R2 show up as an OSPF neighbor on R1? **Yes** 2

Does R2 show up as an OSPF neighbor on R3? **No** 2

What does this information tell you?

Traffic from R3 to the 192.168.2.0/24 network will go through R1. The S0/0/1 interface on R2 is set as passive, so it doesn't send OSPF routing updates. The total cost of 129 is because the traffic from R3 to the 192.168.2.0/24 network goes over two T1 serial links and the R2 GigabitEthernet 0/0 LAN link. 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

```
R2(config)#router ospf 1
R2(config-router)#no passive-interface s0/0/1
R2(config-router)#
01:40:17: %OSPF-5-ADJCHG: Process 1, Nbr 33.33.33.33 on Serial0/0/1 from LOADING to FULL,
Loading Done
```

2

i. Re-issue the **show ip route** command on R3.

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     3.0.0.0/32 is subnetted, 1 subnets
C       3.3.3.3/32 is directly connected, Loopback0
O    192.168.1.0/24 [110/65] via 192.168.13.1, 00:36:28, Serial0/0/0
O    192.168.2.0/24 [110/65] via 192.168.23.1, 00:02:01, Serial0/0/1
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0
     192.168.12.0/30 is subnetted, 1 subnets
O       192.168.12.0/30 [110/128] via 192.168.23.1, 00:02:01, Serial0/0/1
                        [110/128] via 192.168.13.1, 00:02:01, Serial0/0/0
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/30 is directly connected, Serial0/0/0
L       192.168.13.2/32 is directly connected, Serial0/0/0
     192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.23.0/30 is directly connected, Serial0/0/1
L       192.168.23.2/32 is directly connected, Serial0/0/1
```

What interface is R3 using to route to the 192.168.2.0/24 network? **s0/0/1** 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65=64+1, which comes from one T1 serial link (cost=64) plus the R2 g0/0 LAN link (cost=1). 2

Is R2 listed as an OSPF neighbor to R3? **Yes** 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

We need to handle more data with our equipment today. And today's equipment can support link speeds faster than 100 Mb/s. So, to accurately calculate costs for faster links, we should use a higher default reference bandwidth setting.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

```
R1#show ip route ospf
O    192.168.2.0 [110/782] via 192.168.12.2, 00:00:02, Serial0/0/0
O    192.168.3.0 [110/782] via 192.168.13.2, 00:00:02, Serial0/0/1
     192.168.23.0/30 is subnetted, 1 subnets
O       192.168.23.0 [110/845] via 192.168.13.2, 00:00:02, Serial0/0/1
                     [110/845] via 192.168.12.2, 00:00:02, Serial0/0/0
```

1. 192.168.3.0/24: 782, because R1 s0/0/1 + R3 g0/0, which is 781+1=782.

2. 192.168.23.0/30: 845, because R1 s0/0/1 + R3 s0/0/1, which is 781+64=845. 2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

```
R1#show ip route ospf
O    192.168.2.0 [110/782] via 192.168.12.2, 00:11:51, Serial0/0/0
O    192.168.3.0 [110/781] via 192.168.13.2, 00:03:53, Serial0/0/1
     192.168.23.0/30 is subnetted, 1 subnets
O       192.168.23.0 [110/1562] via 192.168.13.2, 00:01:44, Serial0/0/1
                     [110/1562] via 192.168.12.2, 00:01:44, Serial0/0/0
```

ANS: 1562.

Each serial link now has a cost of 781. The route to the 192.168.23.0/24 network goes over two serial links, so the total cost is 1,562 (781 + 781).

2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

```
R1#show ip route ospf
O    192.168.2.0 [110/782] via 192.168.12.2, 00:13:10, Serial0/0/0
O    192.168.3.0 [110/1563] via 192.168.12.2, 00:00:06, Serial0/0/0
     192.168.23.0/30 is subnetted, 1 subnets
O       192.168.23.0 [110/1562] via 192.168.12.2, 00:00:06, Serial0/0/0
```

3

The route to 192.168.3.0/24 on R1 is now going through R2 because OSPF always selects the path with the lowest total cost. After the bandwidth changes, the path R1 → R2 → R3 → 192.168.3.0/24 has a cost of 781 + 781 + 1 = 1563. This is lower than the direct path from R1 to R3, whose cost is 1562 + 10 = 1572. Therefore, R1 prefers the path through R2.

6

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   It is important to control the router ID in OSPF because the router ID uniquely identifies each router in the OSPF domain. It is used in neighbor relationships, LSA generation, and DR/BDR elections. If the router ID changes unexpectedly, OSPF adjacencies may need to be reestablished and troubleshooting becomes more difficult. Therefore, it is best to configure a stable router ID manually or use a loopback interface.

   4

2. Why is the DR/BDR election process not a concern in this lab?

   The selection of DR/BDR takes place exclusively on multi-access networks, such as Ethernet or Frame Relay. Given that the serial connections in this laboratory are point-to-point, there is no requirement for a DR/BDR selection process.

   4

3. Why would you want to set an OSPF interface to passive?

   After configuring the OSPF interface as passive, it will cease transmitting OSPF Hello packets, thereby preventing unauthorized OSPF adjacencies and enhancing data security. Additionally, the necessity for OSPF neighbors is eliminated, which in turn reduces network overhead and improves operational efficiency.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The aspect of OSPF that makes it hierarchical and allows for the creation of scalable networks is its use of "areas." OSPF divides a network into multiple areas, with a backbone area (Area 0) connecting them. This hierarchical structure reduces routing table size and limits the scope of route updates, improving efficiency and scalability.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   Network 10.0.0.0 0.255.255.255 area 0

   Because 10.0.0.0 is the starting IP address of the range and 0.255.255.255 is the wildcard mask, which matches all addresses in the 10.0.0.0/8. Area 0 represents the interfaces belonging to OSPF area 0.

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The router will use the route from EIGRP for the routing decision.

   Because in routing protocols, the lower the administrative distance (AD), the more preferred the route. The Ads for the protocols in this situation are: RIP is 120, OSPF is 110, EIGRP is 90. Since EIGRP has the lowest AD, its route will be selected regardless of its metric value.

   Metrics are only compared when routes come from the same protocol.

   4